

SANSKAR KHANDELWAL

Computer Science student specializing in AI systems and backend engineering

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PROFESSIONAL SUMMARY

Computer Science student specializing in AI systems and backend engineering with experience building containerized microservices, retrieval-based AI platforms, and multi-agent systems. Skilled in Python, FastAPI, Docker, CI/CD pipelines, and distributed data systems with strong foundations in algorithms, operating systems, and database systems.

SKILLS

Programming Languages : Python, Java, SQL, Shell Scripting
Frameworks & Libraries : FastAPI, REST APIs, Microservices
Tools & Platforms : Docker, GitHub Actions (CI/CD)
Databases : PostgreSQL, MongoDB, Redis, FAISS, ChromaDB
Soft Skills : Data Structures & Algorithms, Operating Systems, DBMS, Object-Oriented Programming, Machine Learning, Deep Learning, Statistics

PROJECTS

Hotel Intelligence Agent & MCP Server | 🌐 Jan 2026 - Present

- Built a state-aware AI service using LangGraph to automate MongoDB analytics through backend APIs with cyclic reasoning and query correction.
- Developed a custom Model Context Protocol (MCP) server exposing 20+ tools for schema discovery and aggregation pipelines for automated data analysis workflows.
- Implemented Redis-based persistence enabling sub-50ms state recovery across concurrent API requests.

Technologies / Tools Used : LangGraph, FastAPI, MongoDB, Redis

Pragati AI: Intelligent Multi-Agent Decision Platform Oct 2025 - Present

- Designed an adaptive RAG pipeline with routing strategies for simple and multi-hop reasoning queries.
- Architected a LangGraph-based agent orchestration system coordinating 15+ specialized agents for complex agricultural decision workflows.
- Implemented Redis-based persistence to maintain long-running workflows and session states across reasoning steps.

Technologies / Tools Used : LangChain, RAGAS, FAISS, Docker

WORK EXPERIENCES

AI Project Intern | Team Lead | Internship May 2025 - Jul 2025

Annam.AI — Centre of Excellence, IIT Ropar

Ropar, India

- Led a 5-member engineering team to design and deliver AgroVista, an AI-powered agricultural advisory platform, managing the full software development lifecycle (SDLC) from system design to deployment within 2 months.
- Designed a scalable RAG-based AI pipeline using Sentence Transformers, FAISS, and ChromaDB, integrated with Gemini 1.5 Flash and exposed through FastAPI REST microservices for real-time AI recommendations.
- Developed containerized FastAPI microservices backed by Supabase (PostgreSQL) for secure cloud-hosted data operations and scalable backend services.

Technologies / Skills Used : Python, FastAPI, Docker, Supabase, PostgreSQL, FAISS, ChromaDB

EDUCATIONS

GLA University Aug 2023 - Present

B.Tech in Computer Science (AI/ML & IoT)

CGPA : CPI: 8.46/10

AWARDS & ACHIEVEMENTS

Received 4,00,000 cash prize for developing an AI-based agricultural solution.